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Module 4: Software Design
Session 2 stretch: the ideas behind usable design

Software Developer Level 4



Nielsen's ten usability heuristics

The idea

Ten broad rules of thumb for spotting usability problems, such as visibility of system status, error prevention, and recognition rather than recall.

Several reviewers check a design against them separately, then compare notes.

Why it matters to a developer

They give a team shared words: "this breaks error prevention" is quicker to agree on than "it feels wrong".

A heuristic review is cheap but finds different problems from testing with users, so do both.

Try this: which heuristic does a form break if it clears every answer after one mistake?

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Fitts's law (1954)

POINTING

> **Big targets close to your thumb are quicker and safer to hit.**

The idea: the time to hit a target grows with its distance and shrinks as the target gets bigger.

Try this: on your screen 3, where should the call button go, and how big should it be? Why?

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Hick's law (1952)

More choices, slower decisions

The idea: the time to choose grows with the number of equally likely options, roughly with its logarithm rather than in a straight line.

Why it matters: a long list of repair types slows everyone down, but splitting it into many tiny steps can be slower still.

Try this: Kestrel Moor's data has a category for each kind of repair. Would you show them all on screen 1, or group them? Why?

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WCAG's four principles: POUR

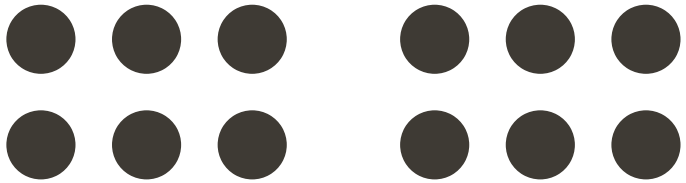
Every WCAG 2.2 success criterion sits under one of four principles.

Principle	In plain words	One check on your wireframe
Perceivable	Everyone can see or hear it in some form	Every icon also has words
Operable	Everyone can use it by touch, keyboard or voice	Large targets; nothing timed
Understandable	People understand it and know what to do next	Plain words; errors say how to fix
Robust	It works with browsers and assistive technology	Real buttons and labels in code

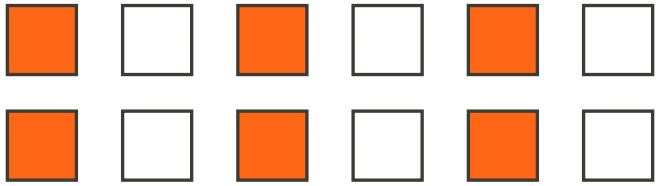
Try this: which principle does each point of Kestrel Moor's standard belong to?

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Gestalt grouping: proximity and similarity



Proximity: two groups of six



Similarity: columns, not rows

The idea

People see things that are close together, or look alike, as one group, before they read a word.

Why it matters

Spacing tells a tenant which label belongs to which box. A label halfway between two boxes belongs to neither.

Try this

On your screen 1, is each label closer to its own box than to the box above it?

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Jakob's law: a heuristic, not a law

1. **The idea**
   People spend most of their time on other sites and apps, so they expect yours to work the same way.

2. **Why it matters**
   Familiar patterns, such as radio buttons or an error summary at the top, need no learning.

3. **The limit**
   It is an observation, not a tested law. Break a convention only when testing shows a better way.

Try this: name one familiar pattern you reused in your prototype, and where it came from.

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Optional self-study for after Session 2. Each of the next six slides gives one idea from usability or accessibility research: the idea in plain words, why it matters to a developer, and a "Try this" question you can apply to Kestrel Moor Homes or to a project of your own.

The notes on each slide repeat the idea, say why it matters and give the full reference, so you can read the original. Where an idea is contested or often over-stated, the notes say so.

Use the slides in any order. They take about 30 minutes to read and try. Nothing in the Consolidate Quiz depends on them. If you bring an answer to Session 3, you may be invited to share it at the start.

If you apply a question to your own project, keep the example general: no personal or customer data, no credentials or confidential material, and no employer data in AI tools your employer has not approved.

Nielsen's ten usability heuristics

The idea

Ten broad rules of thumb for spotting usability problems, such as visibility of system status, error prevention, and recognition rather than recall.

Several reviewers check a design against them separately, then compare notes.

Why it matters to a developer

They give a team shared words: "this breaks error prevention" is quicker to agree on than "it feels wrong".

A heuristic review is cheap but finds different problems from testing with users, so do both.

Try this: which heuristic does a form break if it clears every answer after one mistake?

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The idea: ten broad rules of thumb for spotting usability problems. They are: visibility of system status; match between the system and the real world; user control and freedom; consistency and standards; error prevention; recognition rather than recall; flexibility and efficiency of use; aesthetic and minimalist design; help users recognise, diagnose and recover from errors; and help and documentation.

Why it matters: a team gets shared words for a problem. "This breaks error prevention" is quicker to agree on than "it feels wrong". Several reviewers check a design separately and then compare notes, because each one finds different problems.

Try this: which heuristic does a form break if it clears every answer after one mistake? Think about what the user can and cannot do next.

A caution: these are heuristics, not tested laws. A heuristic review is cheap, but it finds different problems from watching real users, so do both.

Reference: Nielsen, J. (1994) "Enhancing the explanatory power of usability heuristics", Proceedings of ACM CHI '94, pages 152 to 158. Summary: Nielsen Norman Group, "10 Usability Heuristics for User Interface Design" (first published 24 April 1994). The first set came from Molich, R. and Nielsen, J. (1990) "Improving a human-computer dialogue", Communications of the ACM, 33(3), pages 338 to 348.

Fitts's law (1954)

POINTING

> **Big targets close to your thumb are quicker and safer to hit.**

The idea: the time to hit a target grows with its distance and shrinks as the target gets bigger.

Try this: on your screen 3, where should the call button go, and how big should it be? Why?

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The idea: the time it takes to hit a target grows with its distance and shrinks as the target gets bigger.

Why it matters: on a phone, the main action should be large and within easy reach of a thumb, and small targets close together cause mis-taps. The same reasoning sits behind WCAG 2.2's target size criteria (24 by 24 CSS pixels as the AA minimum; about 44 by 44 in the AAA criterion) and behind full-width buttons on mobile forms.

Try this: on your own screen 3, where should the call buttons go, and how big should they be? Think about a tenant who is stressed, moving, or holding the phone in one hand.

A caution: Fitts's experiments had people tapping between metal plates with a stylus, not touch screens. The relationship has been replicated widely for pointing with a mouse and a finger, but the exact numbers depend on the device and the person, so treat it as a strong guide rather than a formula for button sizes.

Reference: Fitts, P. M. (1954) "The information capacity of the human motor system in controlling the amplitude of movement", Journal of Experimental Psychology, 47(6), pages 381 to 391.

Hick's law (1952)

More choices, slower decisions

The idea: the time to choose grows with the number of equally likely options, roughly with its logarithm rather than in a straight line.

Why it matters: a long list of repair types slows everyone down, but splitting it into many tiny steps can be slower still.

Try this: Kestrel Moor's data has a category for each kind of repair. Would you show them all on screen 1, or group them? Why?

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The idea: the time it takes to choose grows with the number of equally likely options, roughly with its logarithm rather than in a straight line. Hick measured how long people took to react when one of up to ten lamps lit.

Why it matters: a long list of repair types slows everyone down, but splitting a list into many tiny steps can be slower still. The number of options is one cost among several.

Try this: Kestrel Moor's data has a category for each kind of repair. Would you show them all on one screen, or group them? What would you test with tenants to find out?

Contested: Hick's law is often quoted as "fewer options is always better". That over-states it. The logarithmic relationship holds for simple choices between familiar, equally likely options. When people scan an unfamiliar list, time tends to grow more steadily with its length, and when they already know what they want, the length of the list matters little. Use it as a prompt to question long lists, not as a rule that menus must be short.

Reference: Hick, W. E. (1952) "On the rate of gain of information", Quarterly Journal of Experimental Psychology, 4(1), pages 11 to 26.

WCAG's four principles: POUR

Every WCAG 2.2 success criterion sits under one of four principles.

Principle	In plain words	One check on your wireframe
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Try this: which principle does each point of Kestrel Moor's standard belong to?

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The idea: every WCAG 2.2 success criterion sits under one of four principles. Perceivable: everyone can see or hear the content in some form. Operable: everyone can use it, by touch, keyboard or voice. Understandable: people understand it and know what to do next. Robust: it works with browsers and assistive technology.

Why it matters: the principles help you decide where a problem belongs and what kind of fix it needs. A low-fidelity wireframe can show many Understandable and Operable decisions; Perceivable and Robust are mostly settled in the visual design and the code (text alternatives, real form controls, labels coded as labels).

Try this: take the seven points of Kestrel Moor's accessibility standard (Case Brief Part 3) and decide which principle each one belongs to. Some could sit under two: which, and why?

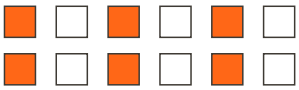
These four principles are the structure of WCAG itself, not a matter of debate. WCAG 3 is in development as a draft; WCAG 2.2 is the current version and the one Kestrel Moor's standard names.

Reference: W3C (2023) Web Content Accessibility Guidelines (WCAG) 2.2, W3C Recommendation, October 2023; and W3C, Understanding WCAG 2.2, which explains each principle and criterion.

Gestalt grouping: proximity and similarity



Proximity: two groups of six



Similarity: columns, not rows

The idea

People see things that are close together, or look alike, as one group, before they read a word.

Why it matters

Spacing tells a tenant which label belongs to which box. A label halfway between two boxes belongs to neither.

Try this

On your screen 1, is each label closer to its own box than to the box above it?

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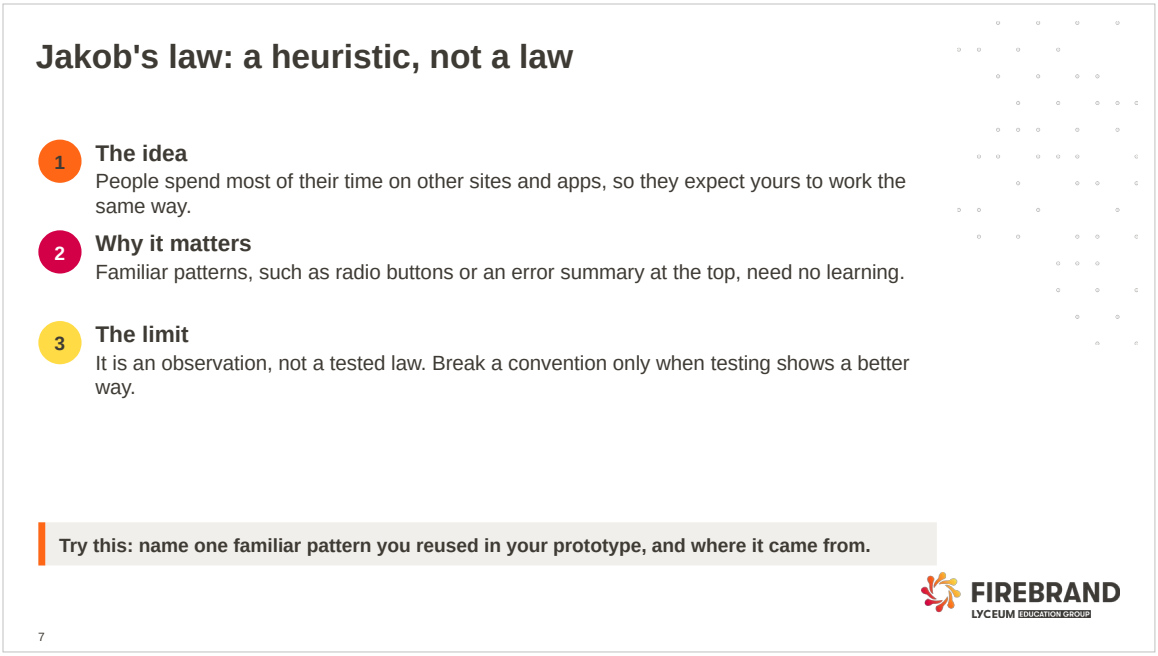
The idea: people see things that are close together, or that look alike, as one group, before they read a word. In the drawing, the dots are identical but the gap in the middle makes two groups (proximity); the squares are evenly spaced but the alternating fill makes columns stand out (similarity). Other grouping principles include common region (things inside one border belong together) and continuity.

Why it matters: spacing tells a user which label belongs to which box. A label halfway between two boxes belongs to neither, and an error message far from its field is easy to miss.

Try this: on your own screen 1, is each label closer to its own box than to the box above it? Is anything that should stand apart, such as the emergency question, set apart by space or a border?

A caution: Gestalt principles describe how perception tends to organise what we see. They are reliable for layout decisions but are not precise laws with numbers attached.

Reference: Wertheimer, M. (1923) "Untersuchungen zur Lehre von der Gestalt II", Psychologische Forschung, 4, pages 301 to 350. English translation: "Laws of organization in perceptual forms", in Ellis, W. D. (ed.) (1938) A Source Book of Gestalt Psychology, London: Routledge and Kegan Paul.



The idea: people spend most of their time on other sites and apps, so they expect yours to work the same way as the ones they already know.

Why it matters: familiar patterns, such as radio buttons for a single choice, an error summary at the top of a form or a Back link at the top left, need no learning. Every place where your design breaks a habit is a place where people have to stop and work it out.

Try this: name one familiar pattern you reused in your prototype and where it came from. Would Kestrel Moor's tenants be likely to know it?

Contested: despite the name, this is a heuristic, not a law. It is an observation about where people's habits come from, not the result of a controlled experiment, and it does not forbid new ideas. Conventions change when enough products adopt a better pattern, so break one only when testing shows a better way.

Reference: Nielsen, J. (2000) "End of Web Design", Alertbox, Nielsen Norman Group, 22 July 2000. Nielsen Norman Group also discusses it under "Consistency and standards" in its article on the ten usability heuristics.